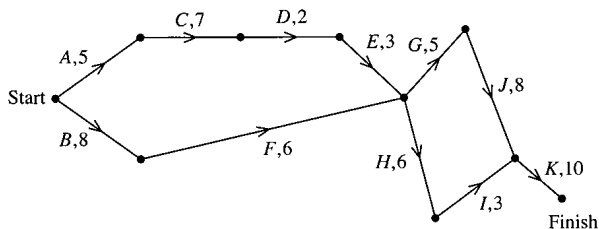


- 1 A project requires completion of 11 tasks $A, B, C, \dots, K$.

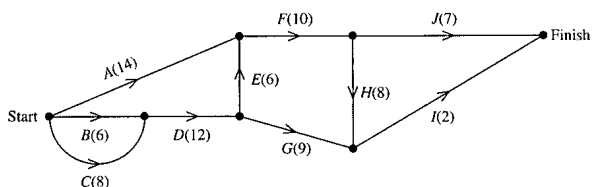
A network diagram for the project giving the completion time for each task, in minutes, is shown.



- Find the minimum time to complete the project. (1 mark) **Hard**
- State the critical path for this project. (1 mark) **Medium**
- A new task, X , is to be added to the project. The earliest starting time for X is 17 minutes, the latest starting time for X is 18 minutes and X has a completion time of 12 minutes. Add task X to the given network diagram above AND state the float time for this task. (2 marks) **Hard**

(Q36, 2021 HSC)

- 2 The preparation of a meal requires the completion of all ten activities A to J . The network diagram shows the activities and their completion times in minutes.

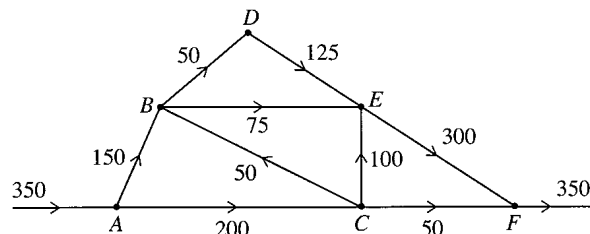


- What is the minimum time needed to prepare the meal? (1 mark) **Hard**
- List the activities which make up the critical path for this network. (2 marks) **Medium**
- Complete the table below, showing the earliest start time and float time for activities A and G . (2 marks) **Medium**

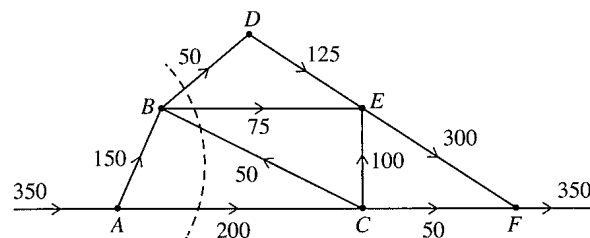
Activity	Earliest start time (minutes)	Float time (minutes)
A		
G		

(Q26, 2020 HSC)

- 3 The network diagram shows a series of water channels and ponds in a garden. The vertices A, B, C, D, E and F represent six ponds. The edges represent the water channels which connect the ponds. The numbers on the edges indicate the maximum capacity of the channels.



- Determine the maximum flow of the network. (2 marks) **Medium**
- A cut is added to the network, as shown.



Is the cut shown a minimum cut? Give a reason for your answer. (1 mark) **Hard**

(Q30, 2020 HSC)

- 4 A project requires activities A to F to be completed. The activity chart shows the immediate prerequisite(s) and duration for each activity.

Activity	Immediate prerequisite(s)	Duration in hours
A	—	2
B	A	6
C	A	5
D	B	2
E	C, D	4
F	E	1

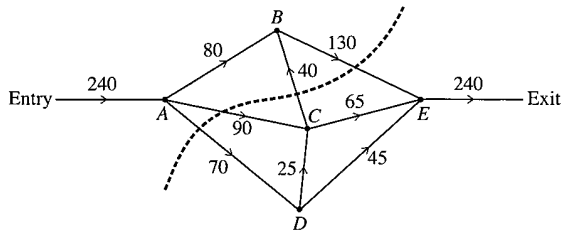
- By drawing a network diagram, determine the minimum time for the project to be completed. (3 marks) **Medium**
- Determine the float time of the non-critical activity. (1 mark) **Hard**

(Q26, 2019 HSC)

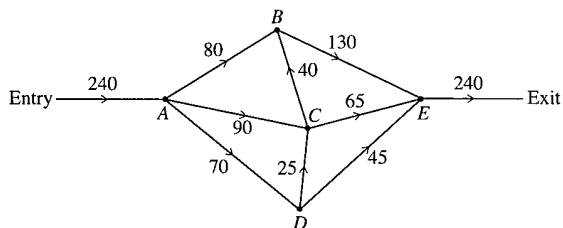
- 5 A museum is planning an exhibition using five rooms.

The museum manager draws a network to help plan the exhibition. The vertices A, B ,

C , D and E represent the five rooms. The numbers on the edges represent the maximum number of people per hour who can pass through the security checkpoints between the rooms.

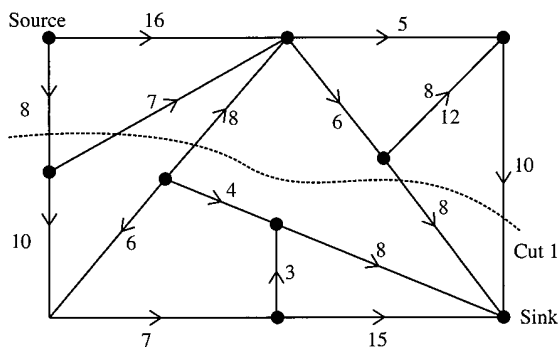


- What is the capacity of the cut shown? (1 mark) **Hard**
- The museum manager is planning for a maximum of 240 visitors to pass through the exhibition each hour. By using the 'minimum cut – maximum flow' theorem, the manager determines that the plan does not provide sufficient flow capacity. Draw the minimum cut onto the network below and recommend a change that the manager could make to one or more security checkpoints to increase the flow capacity to 240 visitors per hour.



(2 marks) **Hard**
(Q40, 2019 HSC)

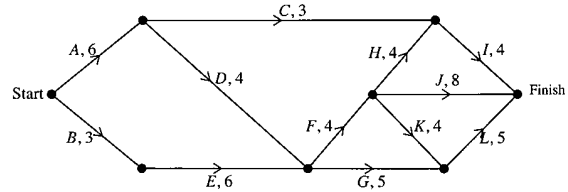
- In the network below, the values on the edges represent maximum possible flow between each pair of vertices. The arrows show the direction of flow. Cut 1 is shown on the network diagram.



What is the capacity of Cut 1? (1 mark) **Easy**

Bonus question (see page iv)

- To restore a vintage motor bike, a group of friends have determined there are 12 activities that need to be completed. The network below shows these 12 activities and their completion times in hours.



- Determine the earliest starting time of activity G . (1 mark) **Medium**
- What is the latest starting time of activity J ? (1 mark) **Medium**
- What is the float time of activity E ? (1 mark) **Medium**

The minimum time in which all 12 activities can be completed is 23 hours.

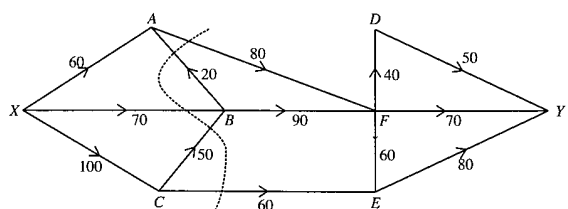
- What is the latest starting time for activity P so as not to increase the minimum completion time of the project? (1 mark) **Hard**

Just before the friends start the restoration they agree to add another activity, P , to the project. Activity P will take 3 hours to complete. Activity P will have no prerequisite activity, but will need to be completed before activity F commences.

- Determine the least cost of this time reduction for activity A to give the greatest reduction in the minimum completion time of the project. (1 mark) **Hard**

Bonus question

- The diagram below shows the flow capacity, in megalitres per hour (ML/h), of a network of gas pipes which connect a gas field (X) to a large storage facility (Y).



- What is the capacity of the cut shown on the diagram? (1 mark) **Easy**
- Find the maximum flow of gas across the network. (1 mark) **Medium**
- To increase the flow of gas, the energy company plans to build a new pipeline

joining *A* to *Y* which will have a capacity of 70 ML/h.

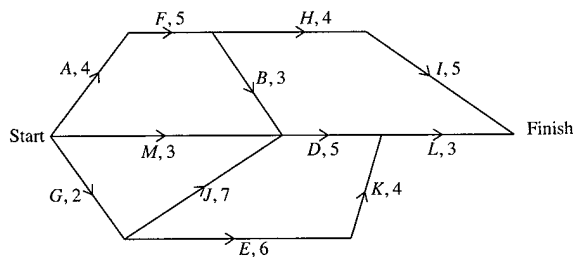
- Draw the new pipeline on the diagram. (1 mark) **Medium**
- What will be the new maximum flow capacity of the network? (1 mark) **Medium**

Bonus question

- 9 A project will be undertaken at a zoo. This project involves the 13 activities shown in the table below. The duration, in hours, and any prerequisite activities are also included in the table.

Activity	Duration	Prerequisites
A	4	–
B	3	F
C	2	H
D	5	J, M, B
E	6	G
F	5	A
G	2	–
H	4	F
I	5	H
J	7	G
K	4	E
L	3	C, D, K
M	3	–

Activity *C* is missing from the diagram for this project which is shown below.



- Complete the activity diagram above by inserting activity *C*. (1 mark) **Medium**
- Determine the earliest starting time for activity *K*. (1 mark) **Medium**
- Given that activity *C* is not on the critical path:
 - write down the activities that are on the critical path in the order they need to be completed. (1 mark) **Medium**
 - find the latest starting time for activity *M*. (1 mark) **Medium**
- James commented: 'If the time allocated to one activity is reduced by one hour then we could complete the project an hour earlier.' Under what circumstances would this occur? (1 mark) **Medium**
- The project manager can choose between two options to complete the project more quickly.

Option 1: The time allocated to activity *D* can be reduced by up to 3 hours at a cost of \$150 per hour.

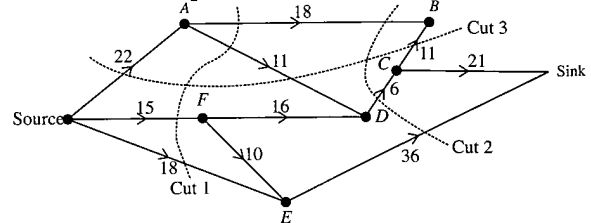
Option 2: The time allocated to activity *I* can be reduced by up to 2 hours at a cost of \$120 per hour.

Which option should she choose?

Support your conclusion with calculations. (2 marks) **Hard**

Bonus question

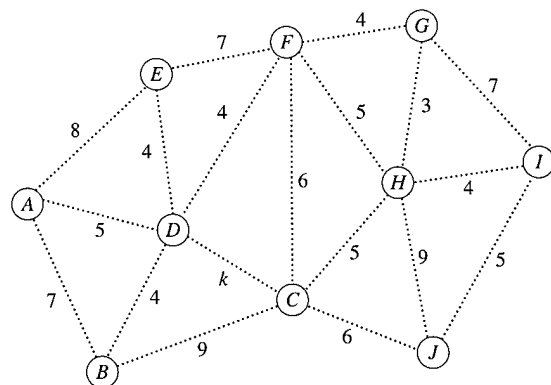
- 10 The network below represents the water flow, in kilolitres per minute, in a series of pipes.



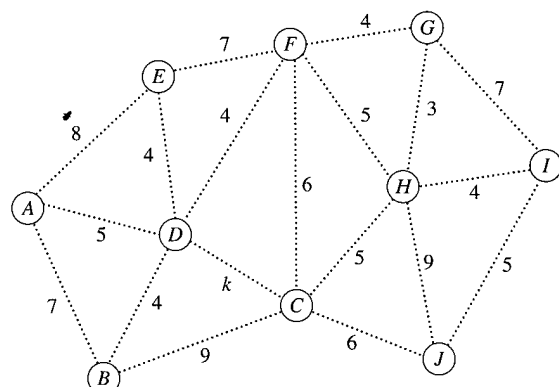
- What is the difference between the flow capacities of Cuts 1 and 2 shown on the diagram? (1 mark) **Medium**
- Explain why Cut 3 is NOT used to calculate the maximum flow. (1 mark) **Easy**
- What is the maximum flow of this network? (1 mark) **Medium**

Bonus question

- 11 The diagram shows a weighted network joining 10 vertices.



- If the minimum spanning tree of the network is 36, show that $k = 3$. (1 mark) **Medium**
- Another copy of the network is provided below for working for this question.



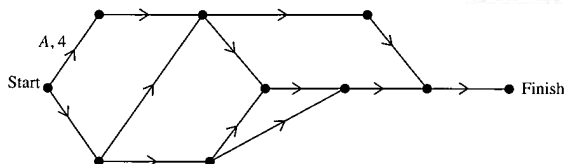
Vertex D is to be removed from the network. What is the weight of the new minimum spanning tree? (2 marks) **Medium**

Bonus question

- 12 A project is comprised of 13 activities and the duration of each activity is measured in days. The table below records the duration of each activity, any prerequisites that exist, the earliest starting time (EST) and the latest starting time (LST) of each activity.

Activity	Duration	Prerequisites	EST	LST
A	4	–	0	2
B	5	–	0	0
C	8	A	4	6
D	6	B	5	8
E	7	B	5	5
F	4	C, D	12	14
G	2	C, D	12	16
H	6	E	12	12
I	3	E	12	19
J	4	G, H	18	18
K	2	I, J	22	22
L	6	F	16	18
M	4	K, L	24	24

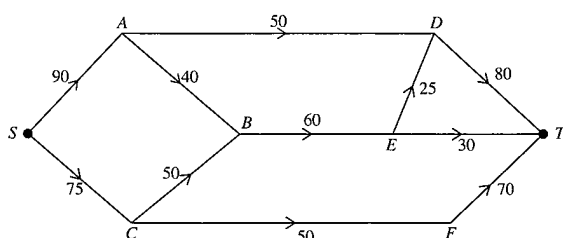
- a What is the float time for activity I ? (1 mark) **Medium**
- b Name the critical path, and hence find the length of the project. (2 marks) **Medium**
- c Use the table to complete the network diagram for the project, recording the duration of each activity. (2 marks) **Medium**



- d Activity P is a new activity to be included in the project. It cannot start until after E is completed and must finish before activity M can commence. The duration of activity P is 18 days. What impact will this new activity have on the completion date of the project? (2 marks) **Hard**

Bonus question

- 13 The diagram shows the traffic flow of vehicles in vehicles/minute, using a road network travelling from a source, S , to a sink, T .



- a Use a dotted line on the diagram to show

the minimum cut and state the maximum flow of vehicles across the road network. (1 mark) **Medium**

- b To increase the maximum flow in the network an engineer decides to upgrade the flow capacity of all roads to a minimum 40 vehicles/minute. The engineer expects this upgrade will improve the capacity of the network by at least 10%. Critically assess the effectiveness of this upgrade. (2 marks) **Hard**

Bonus question

- 14 A burst water main has flooded an underground carpark. The public will be stopped from using the carpark until repairs are made and the area cleaned up.

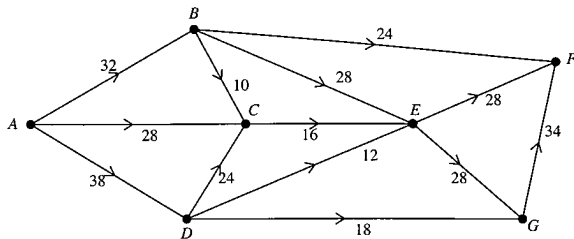
The table below shows the seven activities that need to be completed in order to repair the water pipe and open the carpark. The table also shows the duration for each activity, in minutes, the earliest start times (EST) and the prerequisites for activities.

Activity	Description	Duration	EST	Prerequisite(s)
A	Erect barriers to carpark		0	–
B	Turn off water	5	0	–
C	Pump water from carpark	150	15	A, B
D	Repair water pipe	60	165	C
E	Clean up entire area	45	165	C
F	Turn on water, check for leaks	15	225	D
G	Remove barriers	10		E, F

- a What is the duration of activity A ? (1 mark) **Medium**
- b What is the earliest start time for activity G ? (1 mark) **Hard**
- c Use the table to draw a network diagram for the project. (2 marks) **Medium**
- d What is the duration of the project? (1 mark) **Medium**
- e What is the float time for activity E ? (1 mark) **Medium**
- f It is more complicated cleaning up the area (activity E) than expected as some additional damage was caused by the leak. It will now take twice the time originally planned. Determine the shortest time in which activities A to G can now be completed. (1 mark) **Hard**

Bonus question

- 15 The network diagram shows the flow capacity in kL/min in water pipes connecting A to F.

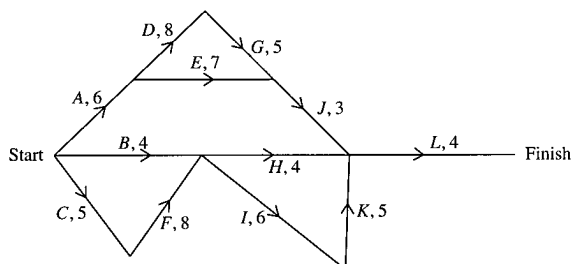


- Identify the sink of this network. (1 mark) **Easy**
- By drawing the minimum cut on the diagram, find the maximum flow of the network. (2 marks) **Medium**
- One pipe in the maximum flow network is to be replaced with a bigger pipe capable of a flow of 40 kL/min. Show that the maximum flow increases by only 8 kL/min. (2 marks) **Hard**

Bonus question

- 16 A new children's playground is to be built in Kahibah. The project involves 12 activities, A to L.

The directed network below shows these activities and their completion times in days.



- Determine the earliest start time for activity J. (1 mark) **Medium**
- The minimum completion time for the playground is 28 days. Write down the critical path for this project. (1 mark) **Medium**
- Which activity has a float time of 7 days? (1 mark) **Medium**
- The completion times for activities D, F, H and K can be reduced by one day. The cost of reducing the completion time by one day for these activities is listed in the table below.

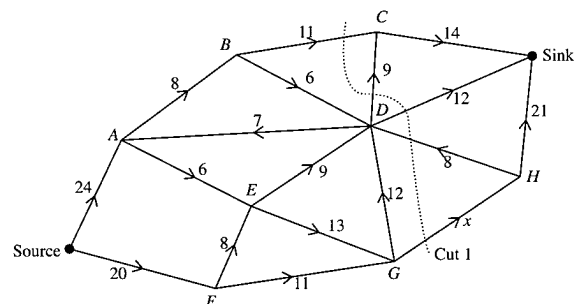
Activity	Cost (\$)
D	4000
F	3000
H	2500
K	2800

What is the minimum additional cost to complete the project in 27 days? (1 mark) **Hard**

- The playground project will be repeated at Redhead, but with the addition of one extra activity, X. The inclusion of this new activity will not extend the project's duration. Activity X will commence once activity I has completed and will finish 2 days before the project is scheduled to be completed. If activity X has a float time of 2 days, what is the duration of the activity? (2 marks) **Hard**

Bonus question

- 17 A flow of oil through pipelines is represented by the network diagram where the direction of the flow (in litres per second) is shown by the arrows.



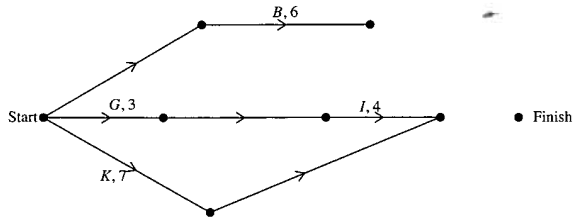
- The line shown as Cut 1 represents a flow of 43 L/second. What is the value of x ? (1 mark) **Easy**
- What is the maximum flow of oil through the network from source to sink? Show working by drawing a line to represent the minimum cut through the network. (1 mark) **Medium**

Bonus question

- 18 The following table, consisting of 12 activities, contains information for a project in a small fabricating workshop.

Activity	Prerequisite(s)	Duration (in hours)
A	–	4
B	A, D	6
C	B, E	3
D	G	4
E	H	5
F	B, E	2
G	–	3
H	G	3
I	H	4
J	F, I, L	4
K	–	7
L	K	5

- Complete the project network below.

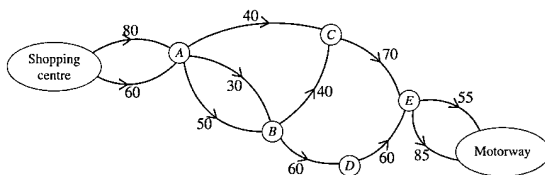


(2 marks) Medium

- b State the critical path and the minimum completion time for this network. (2 marks) Hard
- c Due to unforeseen circumstances, either activity C or activity H will take an extra 4 hours to complete. Which of these activities should be chosen if the completion time for the project is to be kept to a minimum? Justify your answer with calculations. (2 marks) Medium

Bonus question

- 19 The network diagram represents a series of roads connecting a shopping centre to a motorway. Traffic moves in the direction shown on the network. For example, the road from A to C indicates traffic can flow from A to C at a maximum flow of 40 vehicles per minute.



- a What is the existing maximum flow for the road network? (1 mark) Medium
- b It has been decided that one new road will be built to maximise the flow of traffic from the shopping centre to the motorway. This new road cannot start from the shopping centre, and also cannot join the motorway. Which is the best road to build and what will be the minimum capacity of this road to maximise the flow of vehicles across the network? (2 marks) Hard

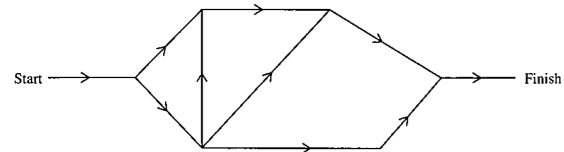
Bonus question

- 20 A construction project involves 10 activities A to J.

The table shows the activities and their duration (in hours), earliest starting time (EST), latest starting time (LST) and prerequisites.

Activity	Duration	EST	LST	Prerequisite(s)
A	6	0	0	–
B	8			A
C	5			A
D	6			B, E
E	7			C
F	9			C
G	5			C
H	6			D, F
I	8			G
J	5			H, I

- a What is the least number of activities that must be completed before activity H can commence? (1 mark) Medium
- b Complete the network diagram for the project.

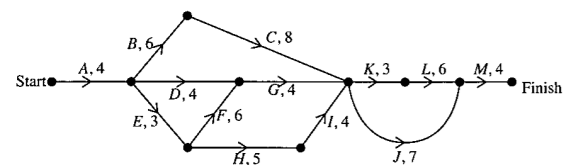


(1 mark) Hard

- c Hence, or otherwise, complete the table. (2 marks) Hard
- d What is the duration of the project? (1 mark) Medium
- e A problem with the project means that activity G will be delayed for 12 hours after activity C has been completed. If any delay in the project costs \$1500 per hour, what will be the additional cost for the project? (1 mark) Hard

Bonus question

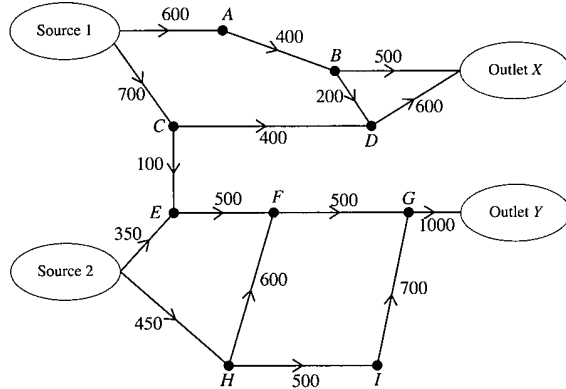
- 21 A project manager outlines a project, in days, using the network diagram below.



- a Use the diagram to find the earliest and latest starting times for activity G. (2 marks) Medium
- b What is the minimum time to complete the project? (1 mark) Medium
- c The project manager decides to subcontract part of the project work. The subcontractors are able to speed up the project to enable an earlier completion time. The activities that can be reduced in time are C, J and L and the maximum reduction for each activity is 2 days. The cost of reducing the time of any activity is \$14500 per day. Determine the minimum time, in days, for the project to be completed and its associated minimum additional cost. (3 marks) Hard

Bonus question

- 22 Stormwater enters a network of pipes at either Source 1 or Source 2 and flows into the ocean through either Outlet X or Outlet Y. The network diagram shows a series of pipes and the arrows represent the flow of stormwater. The numbers next to the arrows represent the maximum rate of stormwater, in kL/minute.



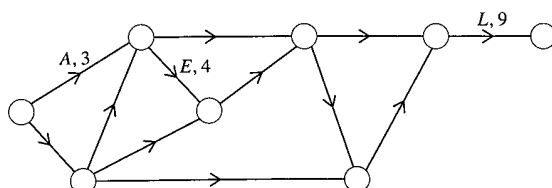
- Explain why water from Source 2 cannot flow to Outlet X. (1 mark) **Easy**
- What is the maximum flow into Outlet Y? (1 mark) **Medium**
- The pipe joining C to E is to be replaced with a larger pipe to increase the flow of stormwater from Source 1. The cost of the new pipe rises with an increase in flow capacity. What should be the flow capacity of the new pipe to minimise cost but maximise flow through Outlet Y? (2 marks) **Hard**

Bonus question

- 23 The table lists the 12 activities in a project.

Activity	Time (in days)	Prerequisite(s)
A	3	-
B	3	-
C	1	B
D	8	A, C
E	4	A, C
F	6	B
G	11	B
H	4	E, F
I	7	D, H
J	11	D, H
K	9	G, I
L	9	J, K

- Use the table to complete the network diagram below.

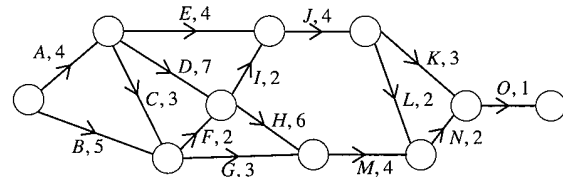


- What is the float time for activity J? (1 mark) **Medium**

- Name the critical path. (1 mark) **Medium**
- If the earliest date that activity F can start is 17 July, what is the latest starting date for activity I? (1 mark) **Medium**

Bonus question

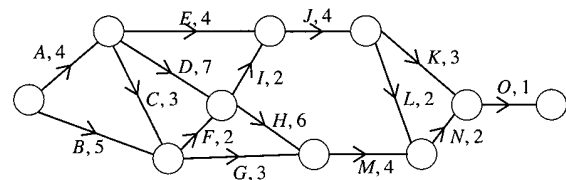
- 24 The diagram shows activities in a project.



- Complete the activity chart for the network diagram above. (2 marks) **Medium**

Activity	Time (in days)	Prerequisite(s)
A	4	-
B	5	-
C	3	A
D	7	A
E	4	A
F	2	
G	3	B, C
H		
I	2	D, F
J	4	E, I
K	3	J
L	2	J
M	4	G, H
N	2	L, M
O	1	

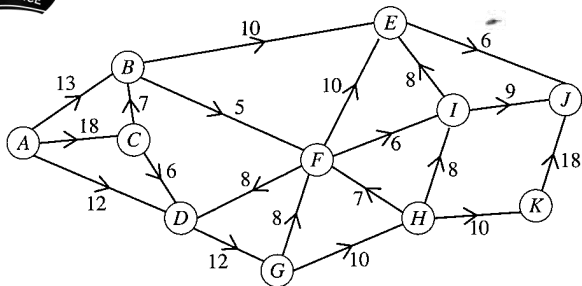
- Identify the critical path and find its length. (2 marks) **Hard**
- What is the slack time for activity L? (1 mark) **Medium**
- The time for activity H can be reduced by up to 4 days at a cost of \$6000 per day. What is the total cost of minimising the project's duration? Support your explanation with calculations. There is another network diagram below to assist with your solution. (2 marks) **Hard**



Bonus question

- 25 The diagram shows a network of water pipes where the rate of flow is represented on each length of pipe, in kL/minute. The direction of flow is shown by the direction of the arrow.

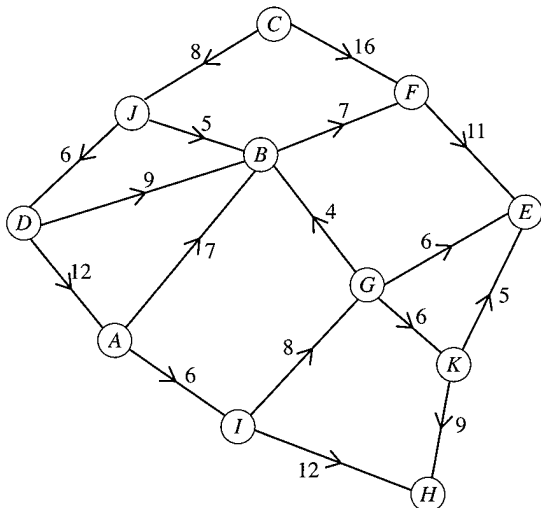
Critical Path



- Draw the line of minimum cut on the diagram. What is the maximum flow of water from A to J? (2 marks) **Medium**
- An additional pipeline is to be constructed from B to J. Water will flow through this pipe to J at the rate of 20 kL/minute. What is the increase in the maximum flow through the network? (1 mark) **Hard**

Bonus question

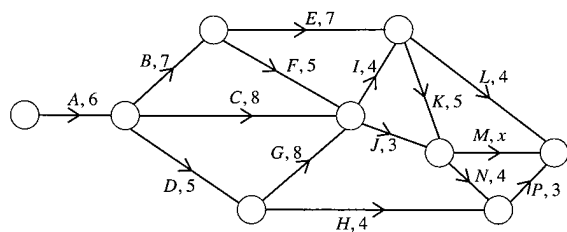
- 26 The diagram represents a network of sewer pipes where capacity is measured in litres per second and the direction of flow is represented by arrows.



Supposing D is the source and E is the sink for the flow across the network, draw a minimum cut and calculate the maximum flow across the network. (1 mark)

Bonus question **Medium**

- 27 To revitalise a children's playground area, a project officer has determined 15 activities that need to be completed. The network diagram below shows these activities and their completion times in days. The minimum time in which all activities can be completed is 36 days.



- The critical path is ADGIKM. Explain why $x = 8$. (1 mark) **Medium**
- What is the latest starting time for activity I? (1 mark) **Medium**
- What is the float time of activity H? (1 mark) **Medium**
- The time allocation for activity M can be reduced by up to 3 days at a cost of \$2300 per day. Determine the least cost of this time reduction for activity M to give the greatest reduction in the minimum completion time of the project. (1 mark) **Hard**

Bonus question

Critical Path

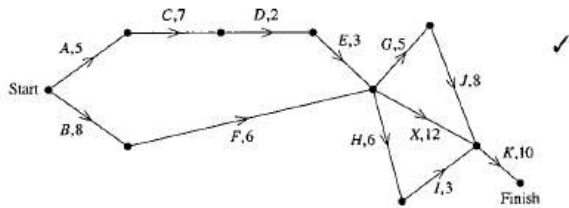
- 1 a Minimum time
 $= (5 + 7 + 2 + 3 + 5 + 8 + 10) \text{ min}$
 $= 40 \text{ min}$

(1 mark)

- b The critical path is A, C, D, E, G, J, K.

(1 mark)

- c EST = 17 min
 So X should start after task E.
 LST = 18 min
 Completion time = 12 min
 Latest finish time = $(18 + 12) \text{ min}$
 $= 30 \text{ min}$
 So X should finish before task K.



The float time is 1 minute.

(2 marks)

- 2 a $8 + 12 + 6 + 10 + 8 + 2 = 46$
 Minimum time = 46 minutes

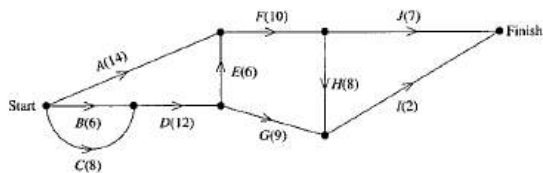
(1 mark)

- b Critical path has activities C, D, E, F, H and I. ✓✓

(2 marks)

Activity	Earliest start time (minutes)	Float time (minutes)
A	0	12
G	20	15

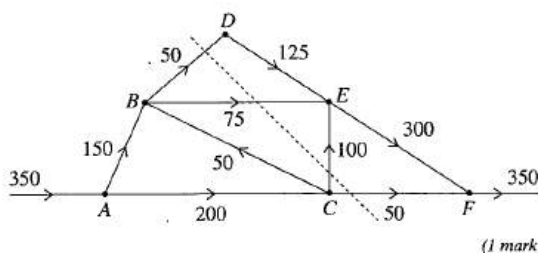
✓✓ (2 marks)



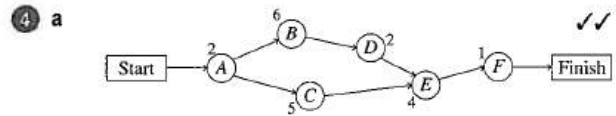
- 3 a Maximum flow = 275
 Vertex B has input of 200 but output of 125. ✓
 Vertex D has input of 50 so can only have output of 50, not 125. ✓
 So vertex E has input and output of 225.

(2 marks)

- b No, the value of the cut is $50 + 75 + 200 = 325$.
 As this is more than the maximum flow it cannot be the minimum cut.
 The minimum cut is:



(1 mark)



Minimum time = 15 hours

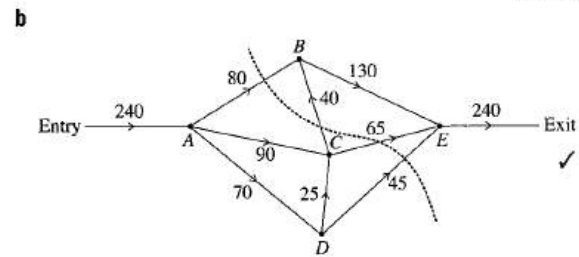
(3 marks)

- b Float time of activity C is 3 hours.

(1 mark)

- 5 a Capacity = $70 + 90 + 130$
 $= 290$

(1 mark)



Increasing the capacity of edge CB to 50 would increase the flow capacity to 240 visitors per hour.

(2 marks)

- 6 $8 + 8 + 10 = 26$

(1 mark)

- 7 a $6 + 4 = 10$
 $\therefore 10 \text{ hours}$

(1 mark)

- b $23 - 8 = 15$
 $\therefore 15 \text{ hours}$

(1 mark)

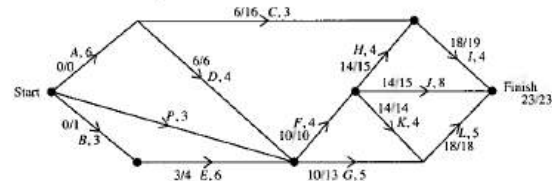
- c $10 - 6 - 3 = 1$
 $\therefore 1 \text{ hour}$

(1 mark)

- d $6 + 4 - 3 = 7$
 $\therefore 7 \text{ hours}$

(1 mark)

- e Path BEFKL is already 22 hours, which means the least cost is \$120.



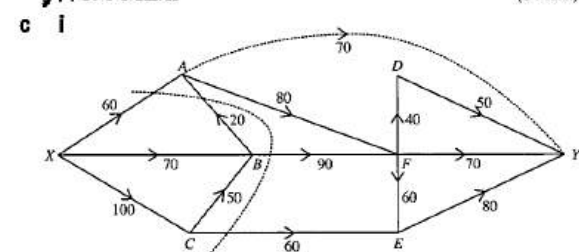
(1 mark)

- 8 a $80 + 70 + 50 + 60 = 260$
 $\therefore 260 \text{ ML/h}$

(1 mark)

- b $40 + 70 + 80 = 190$ (cutting DF, FY and EY)
 $\therefore 190 \text{ ML/h}$

(1 mark)

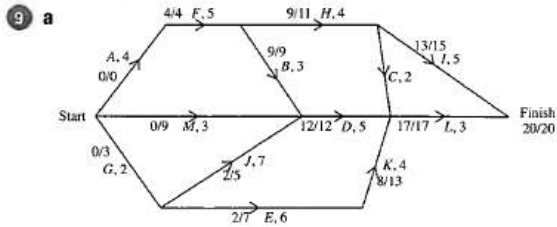


(1 mark)

Critical Path

- ii The new minimum cut is $60 + 90 + 60 = 210$ (cutting AX, AB, BF and CE).
 $\therefore 210 \text{ ML/h}$. (Note: '20' on AB is not used as the arrow is in the reverse direction, from 'sink' to 'source'.)

(1 mark)



- b $2 + 6 = 8$
 $\therefore 8 \text{ hours}$

(1 mark)

- c i A, F, B, D, L

(1 mark)

- ii $12 - 3 = 9$
 $\therefore 9 \text{ hours}$

(1 mark)

- d This would only occur if the activity was on the critical path: A, F, B, D, L .

(1 mark)

- e $AFHI$ presently has a length of 18 hours.
 Option 1: As $2 \times \$150 = \300 , the project time is $20 - 2 = 18 \text{ hours}$ for critical path $AFBDL$.
 Reducing activity D by 2 hours means the two critical paths are now 18 hours each.

(2 marks)

- 10 a Cut 1 is 62 kL/min and Cut 2 is 60 kL/min. The difference is 2 kL/min.

(1 mark)

- b Cut 3 does not separate sink and source.

(1 mark)

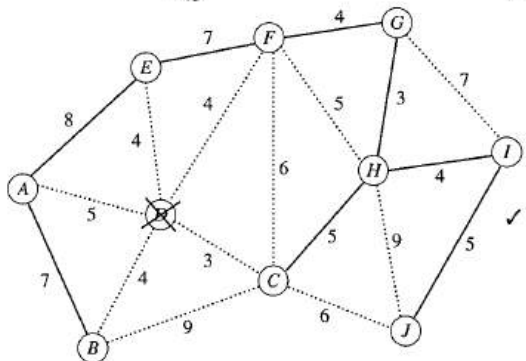
- c $18 + 6 + 10 = 34$
 (the cut goes through Source: E, EF, DC, CB)
 $\therefore$ maximum flow is 34 kL/min.

(1 mark)

- 11 a $5 + 4 + 4 + k + 4 + 4 + 3 + 4 + 5$
 $= 36$
 $33 + k = 36$
 $k = 3$

(1 mark)

- b New weight = $7 + 8 + 7 + 4 + 3 + 5 + 4 + 5$
 $= 43$



(2 marks)

- 12 a $19 - 12 = 7$

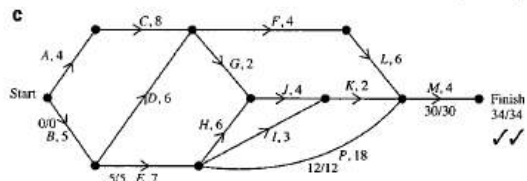
$\therefore$ The float time is 7 days.

(1 mark)

- b $BEHJKM$, a length of 28 days

✓✓

(2 marks)

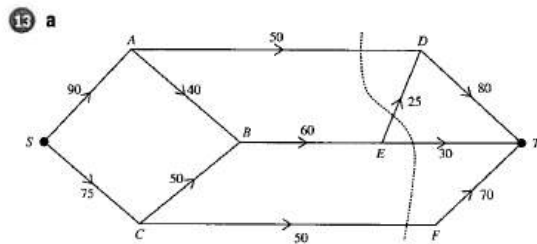


(2 marks)

- d The new critical path is $BEPM$ which takes 34 days. The project will now take 6 days longer to complete.

✓✓

(2 marks)



Maximum flow is
 $50 + 25 + 30 + 50 = 155$
 $\therefore 155 \text{ vehicles/minute}$

(1 mark)

- b Increasing ED and ET to 40 vehicles/minute means that the cut used in part a gives 180 vehicles/minute. But another cut through AD, BE and CF gives 160 vehicles per minute. This is only a 5 vehicles/minute improvement.

As $\frac{5}{155} \times 100\% = 3.2\%$, this is much lower than the engineer's expectations.

✓✓

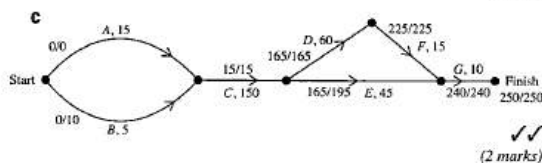
(2 marks)

- 14 a 15 minutes

(1 mark)

- b $225 + 15 = 240$
 $\therefore 240 \text{ minutes}$

(1 mark)



✓✓

(2 marks)

- d $225 + 15 + 10 = 250$
 $\therefore 250 \text{ minutes}$

(1 mark)

- e $250 - 10 - 45 - 165 = 30$
 $\therefore 30 \text{ minutes}$

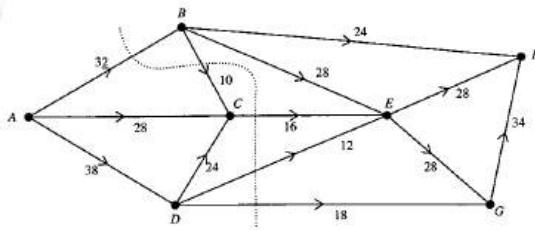
(1 mark)

- f An extra 15 minutes as the critical path is now $ACEG$ rather than $ACDFG$.
 The shortest time is now 265 minutes.

(1 mark)

Critical Path

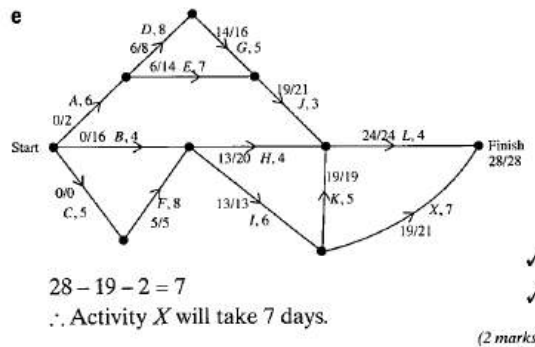
15



- a **F** (1 mark)
- b $32 + 16 + 12 + 18 = 78$ (10 is not used as wrong direction)
 $\therefore 78 \text{ kL/min}$ (2 marks)
- c Replacing CE with a new pipe means the previously calculated maximum flow is now 102 kL/min. However, the new minimum cut through BF, EF and GF gives a maximum flow of $24 + 28 + 34 = 86$.
 As $86 - 78 = 8$, the maximum flow only increases by 8 kL/min. (2 marks)

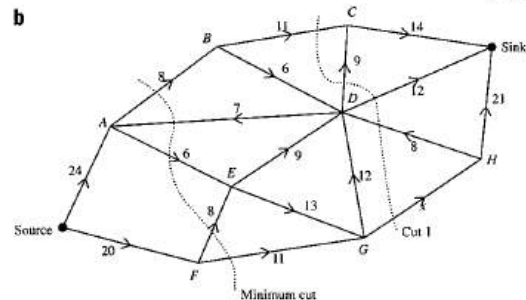
16

- a 19 days (1 mark)
- b **CFIKL** (1 mark)
- c Activity **H** (1 mark)
- d F and K are on the critical path, so one day's reduction would cost \$2800 (by reducing K by one day). (1 mark)



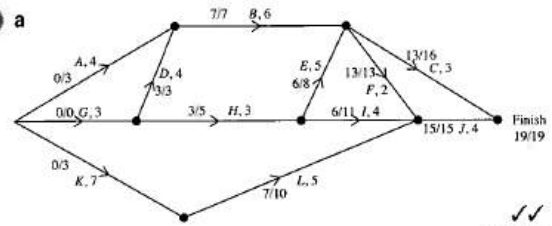
17

- a $11 + 9 + 12 + x = 43$
 $\therefore x = 11$ (1 mark)



The maximum flow is $8 + 6 + 8 + 11 = 33$.
 $\therefore 33 \text{ L/second}$ (1 mark)

18



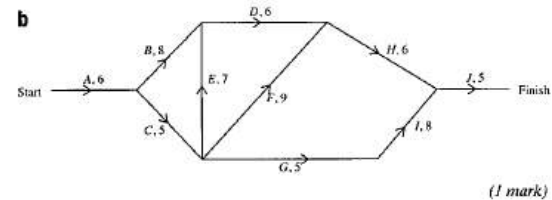
- b **GDBFJ**, 19 hours (2 marks)
- c Increasing C by 4 hours means the path involving C has a maximum of 20 hours (**GDBC**). Increasing H by 4 hours means the path involving H has a maximum of 21 hours (**GHEFJ**). Best to choose activity C to increase the project by just 1 hour. (2 marks)

19

- a Minimum cut:
 $40 + 30 + 50 = 120$
 $\therefore$ maximum flow is 120 vehicles per minute. (1 mark)
- b As $80 + 60 = 140$ and $55 + 85 = 140$, the flow capacity from the source to the sink is 140. Building a new road linking A to E with a capacity of 20 vehicles per minute will ensure a maximum flow of 140 vehicles per minute. (2 marks)

20

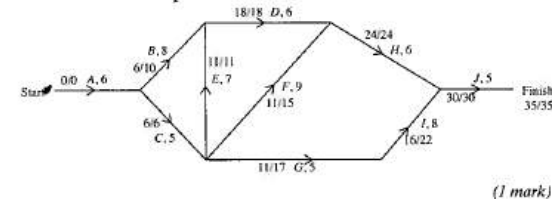
- a Six activities must be completed (**A, B, C, D, E, F**). (1 mark)



c

Activity	Duration	EST	LST	Prerequisite
A	6	0	0	—
B	8	6	10	A
C	5	6	6	A
D	6	18	18	B, E
E	7	11	11	C
F	9	11	15	C
G	5	11	17	C
H	6	24	24	D, F
I	8	16	22	G
J	5	30	30	H, I

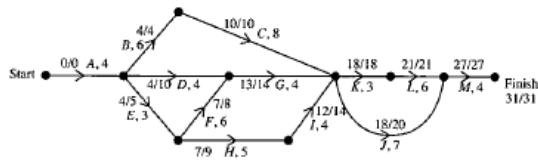
- d The critical path is **ACEDHJ** which is 35 hours. (2 marks)



- e $6 + 5 + 12 + 5 + 8 + 5 = 41$ which means a new critical path. The 6-hour delay will cost $6 \times \$1500 = \9000 . (1 mark)

Critical Path

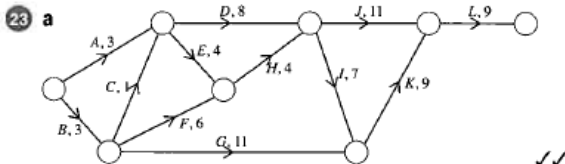
- 21 a EST = 13 days and
LST = 14 days



- b $ABCKLM = 31$ days

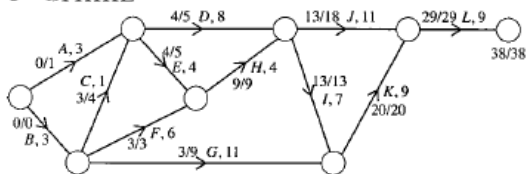
- c Reducing C and L by 2 days each means the $ABCKLM$ path is 27. However, this also means that the $AEFGKLM$ path is 28. The cost of the reduction by 3 days is $3 \times \$14\,500 = \$43\,500$.

- 22 a The arrow on CE means that water can only flow from Source 1.
b $100 + 350 + 450 = 900$
 $\therefore 900 \text{ kL/min}$
c Increase capacity of CE to 150 means the minimum cut and hence maximum flow is 950 kL/min through Outlet Y .
 $\therefore$ the new flow capacity should be 150 kL/min



- b 5 days

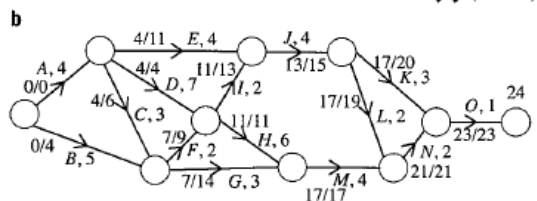
- c $BFHIKL$



- d Activity I lies on the critical path. As $17 - 3 = 14$, the project started on 14 July, and as $14 + 13 = 27$, the latest starting date for activity I is 27 July.

- 24 a

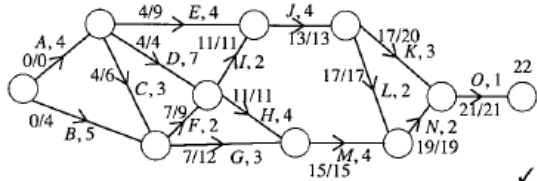
Activity	Time (in days)	Prerequisite(s)
A	4	-
B	5	-
C	3	A
D	7	A
E	4	A
F	2	B, C
G	3	B, C
H	6	D, F
I	2	D, F
J	4	E, I
K	3	J
L	2	J
M	4	G, H
N	2	L, M
O	1	K, N



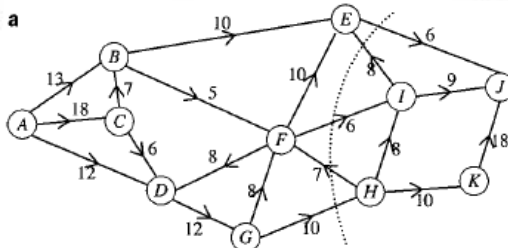
$ADHMNO$, 24 days

- c 2 days

- d Reducing activity H by 2 days means there are now two critical paths.
As $6000 \times 2 = 12\,000$, the cost is \$12 000 to reduce the project's duration by 2 days.



- 25 a

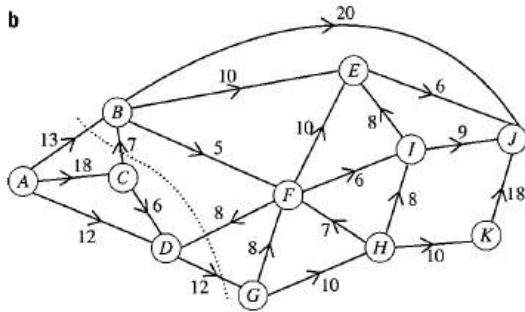


Using the weights for the edges in the direction of Source to Sink:

$$\text{Flow} = 6 + 0 + 6 + 0 + 10 = 22$$

$\therefore$ the maximum flow is 22 kL/min.

Critical Path



New maximum flow

$$= 13 + 7 + 12$$

$$= 32$$

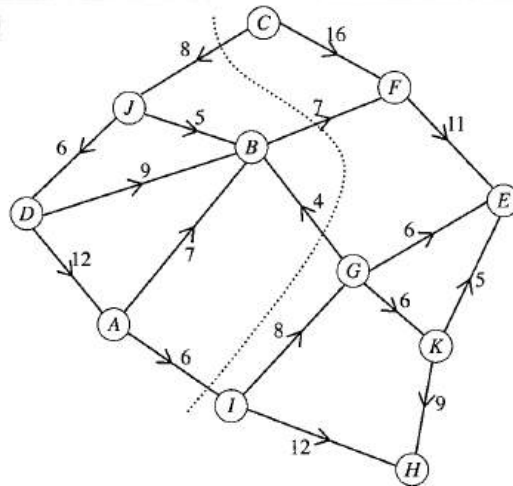
$$\text{Change} = 32 - 22$$

$$= 10$$

$\therefore$ the maximum flow is increased by 10 kL/min.

(1 mark)

26



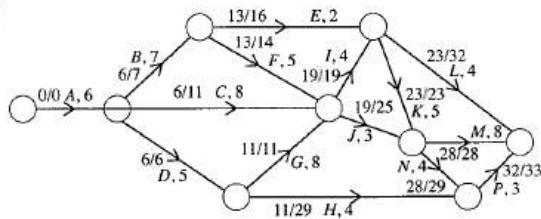
$$\text{Minimum cut} = 7 + 6$$

$$= 13$$

$\therefore$ the maximum flow is 13 L/second.

(1 mark)

27



a $6 + 5 + 8 + 4 + 5 + x = 36$

$$28 + x = 36$$

$$x = 8$$

(1 mark)

b 19 days

(1 mark)

c $29 - 11 = 18$

$\therefore$ 18 days

(1 mark)

d As activity M takes 8 days and activities N + P take 7 days, the manager should only reduce M by 1 day. This means a cost of \$2300.

(1 mark)